Colter Snyder

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Hiring Team
United Launch Alliance
Centennial, CO 80112

Re: Ground Software Development Engineer 2, Ground Software Modernization (Req. 2251)

Dear Hiring Team,

More than 155 consecutive successful missions is a record built on software that people can trust. I would like to help modernize the ground systems behind that record. I am a software engineer at Innovative Scientific Solutions, where I develop ITAR- and CUI-controlled nuclear weapons effects simulation platforms for defense organizations. With an M.S. in Computer Science from the Colorado School of Mines, I would bring hands-on experience from a field where correctness is not optional.

Much of my current work is modernization. I port legacy physics models from Fortran and Python to Rust, keeping their numerical behavior intact while making the code easier to maintain and extend. On a small agile team, I work across the full lifecycle: refining requirements, writing object-oriented designs, UML diagrams and design documentation, implementing features in Python and Rust on Linux, and verifying them with automated tests in GitLab CI/CD pipelines with SonarQube analysis. As an intern at Tyler Technologies, I migrated a legacy VB6 application to C# and overhauled a 2,000-test suite, restoring about 1,500 tests to passing so the team could rely on test-driven development again.

I am glad to see ULA embracing AI-assisted development. I use tools such as Claude Code and Cursor regularly, and I use them with export control in mind. Commercial AI assistants send prompts and code to infrastructure outside an accredited boundary, so at ISSI I use them only for non-CUI, non-ITAR idea exploration and research, and never with controlled code or data. Bringing these tools to bear on controlled work calls for an authorized environment, such as models hosted through AWS Bedrock in GovCloud, and I would be excited to help use them productively within those boundaries at ULA.

My C++ comes mainly from academia: graduate computer graphics coursework, and a semester co-teaching CSCI 220: Data Structures and Algorithms in C++ to 120 students. Teaching that course, and another section of 90, taught me how to explain technical ideas clearly and help less-experienced engineers grow, which is the kind of mentoring this role asks for. Outside work, I have long been drawn to the point where software meets hardware. I design FPGA logic in Verilog and write 6502 and Z80 assembly, and that is a big part of why ground software, which connects code to real flight hardware, appeals to me.

Space has fascinated me for as long as I can remember, and the chance to contribute to it here in Colorado, close to my home in Lakewood, is an opportunity I would be proud to take. I would welcome the chance to discuss how my experience in mission-critical, export-controlled software development can support ULA's Ground Software Modernization team. Thank you for your time and consideration.

Sincerely,

Colter Snyder